JOSHUA MENDOZA

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Skills / Completed Training

SolidWorks, AutoCAD, MasterCAM, GD&T, MIG & TIG Welding, Metal Fabrication, 3D Printing, CNC Machining, Milling, Laser Cutting, Java, MATLAB, Arduino

Education

Bachelor of Science — **California State University, LA** — *Los Angeles, CA* — **Anticipated Graduation: 05/2027**

- Major in Mechanical Engineering
  - Select Coursework: Fundamentals of Vehicle Design, Robotics, Machine Design, Heat Transfer, Experimental Statistics, Fluid Mechanics, Kinematics of Mechanisms, Numerical Methods, Strength of Materials, Thermodynamics
  - Activities: Suspension Lead / Co-Captain of Baja SAE @ CSULA, Participated as an ASME Mechanical Engineer @ CSULA

Engineering Experience

Manufacturing Intern — **Senior Aerospace SSP** — *Burbank CA, USA* — **05/2026-Current**

- Applied GD&T, dimensional tolerancing, and engineering drawing interpretation to support real-time production decisions on precision aerospace hardware.
- Analyzed CMM inspection results and fixture verification data to **identify out-of-tolerance conditions** and determine whether discrepancies originated from part geometry, manufacturing processes, or inspection setup.
- Supported production of Inconel and other high-performance aerospace components, evaluating NCMs, **managed over $2.7 million dollars** in manufacturing orders.
- Investigated dimensional and manufacturing discrepancies across machined and formed components, correlating inspection results with drawing requirements to **determine root cause, manufacturability, and acceptable corrective action**.

Suspension / Co-Captain — **CSULA Baja SAE** — *Los Angeles, CA, USA* — **2025-Current**

- Engineered and fabricated a double A-arm suspension system, **reducing component weight by 3%** while maintaining a safety factor of 1.2 under simulated loading conditions.
- Validated suspension performance through finite element analysis (FEA) in SolidWorks Simulation, reducing **stress concentrations and increasing fatigue life by 12%**.
- Modeled and refined suspension assemblies in SolidWorks, implementing iterative design changes that improved handling stability and vehicle compliance.
- Prototyped components with 3D printing to verify fit, clearance, and kinematics prior to fabrication, **reducing rework time by 2 weeks**.
- Directed cross-functional vehicle integration with chassis, drivetrain, and steering teams to resolve packaging constraints and ensure compatibility between major vehicle systems.
- Presented engineering documentation and simulation data to faculty advisors and SAE judges, earning recognition for technical rigor.

Product Development Intern — **Silver Rainbow** — *Glendale, CA, USA* — **11/2024-03/2026**

- Spearheaded the design and rapid prototyping of proprietary motorized inflator systems by leveraging CAD modeling, 3D printing, and additive manufacturing to enhance airflow efficiency, structural robustness, and user ergonomics.
- Optimized housing architecture through advanced material selection and thermal analysis, improving heat dissipation, structural reliability, and overall system durability during extended operation, **lowering motor operating temperature by 9°C**
- Conducted airflow and pressure testing to validate nozzle designs, achieving a 9% improvement in volumetric flow rates and minimizing backpressure losses, **reducing inflation time by 0.3 seconds**

Projects

Fabricator / Owner — **Chilly Fabrication** — *Los Angeles, CA, USA* — **09/2024-Current**

- Engineer and fabricate custom suspension and tube-chassis components for off-road truck applications, developing solutions around vehicle-specific geometry, packaging, strength, and clearance requirements.
- Measure and optimize suspension and steering geometry, evaluating caster, camber, toe, ride height, shock positioning, bump/droop travel, and component clearances during fabrication and final setup.

ROV Rover — **Cal State LA ROV Challenge** — *Los Angeles, CA, USA* — **08/2022-12/2022**

- Engineered a fully submersible, wired PVC-based ROV with custom waterproof motor housings and buoyancy control, validated through underwater testing.
- Optimized motor mount configurations and propulsion system, **improving power efficiency by 20%**, and secured a Top 3 placement among 10 teams for durability and technical precision